Highlights

Spatiotemporal Dynamics of Conway's Game of Life in the Entropy–Complexity Plane

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- Ordinal patterns reveal the dynamics of the Game of Life.
- Different initial densities lead to distinct but convergent steady states.
- The framework captures transient and long-term spatiotemporal organization.
- Peano-Hilbert curve encoding maps Game of Life dynamics onto the $H \times C$ plane for spatiotemporal complexity analysis
- Exponential relaxation model quantifies convergence rate and half-life, validated across lattice size and bit-depth